

Numerical Analysis homework 1

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I

a

Hermite interpolation: $y(x) = p_3(x)$ So,

$$\begin{array}{c|cccc} -1 & y(-1) & & & \\ 0 & y(0) & y(0) - y(-1) & & \\ 0 & y(0) & y'(0) & y'(0) - y(0) + y(-1) & \\ 1 & y(1) & y(1) - y(0) & y(1) - y(0) - y'(0) & \frac{1}{2}(y(1) - 2y'(0) - y(-1)) \end{array}$$

$$p_3(x) = y(0) + y(0) - y(-1)(x+1) + (y'(0) - y(0) + y(-1))(x+1)x + \frac{1}{2}(y(1) - 2y'(0) - y(-1))x^2(x+1) + \frac{y^4(\xi)}{4!}x^2(x+1)(x-1),$$

$$\int_{-1}^1 y(t) dx = \frac{1}{3}(y(-1) + 4y(0) + y(1)) + \int_{-1}^1 \frac{y^4(\xi)}{4!}x^2(x+1)(x-1) dx$$

this is Simpson's rule.

b

According to the discussion above.

$$E^S(y) = \int_{-1}^1 \frac{y^4(\xi)}{4!}x^2(x+1)(x-1) dx \quad (1)$$

$$= \frac{y^4(\eta)}{4!} \int_{-1}^1 x^2(x+1)(x-1) dx \quad (2)$$

$$= \frac{y^4(\eta)}{4!} \cdot -\frac{4}{15} \quad (3)$$

$$= -\frac{y^4(\eta)}{90}. \quad (4)$$

c

According to (a), (b). Set $x_k = a + kh, h = \frac{b-a}{n}, n \in 2\mathbb{N}$.

$$\int_{-1}^1 y(t) dx = \sum_{k=0}^{n/2} \int_{x_{2k}}^{x_{2k+2}} y(t) dt \quad (5)$$

$$= \sum_{k=0}^{n/2} \int_{x_{2k}}^{x_{2k+2}} p_3(y; x_{2k}, x_{2k+1}, x_{2k+1}, x_{2k+2}; t) + E_k^S(y) dt \quad (6)$$

$$= \sum_{k=0}^{n/2} \frac{1}{3}(y(x_{2k}) + 4y(x_{2k+1}) + f(x_{2k+2})) + \sum_{k=0}^{n/2} E_k^S(y) \quad (7)$$

$$(8)$$

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Consider $\sum_{k=0}^{n/2} E_k^S(y)$, from the discussion in b, we know

$$E_k^S(y) = \int_{x_{2k}}^{x_{2k+2}} \frac{y^4(\eta_k)}{4!} (x - x_{2k+1})^2 (x - x_{2k}) (x - x_{2k+2}) dx = -\frac{y^4(\eta_k)h^5}{90}.$$

So,

$$\sum_{k=0}^{n/2} E_k^S(y) = \sum_{k=0}^{n/2} -\frac{y^4(\eta_k)h^4}{90} \quad (9)$$

$$= -\frac{y^4(\eta)h^4}{90} \cdot \frac{n}{2} \quad (10)$$

$$= -\frac{y^4(\eta)h^4(b-a)}{180} \quad (11)$$

$$(12)$$

II

a

Prove:

The remainder of composite trapezoidal rule is $E_n(f) = -\frac{(b-a)^3}{12n^2} f''(\xi)$.

$f(x) = e^{-x^2}$, $f'(x) = -2xe^{-x^2}$, $f''(x) = (4x^2 - 2)e^{-x^2}$, So $f''(\xi) \leq 2 \Rightarrow |E_n(f)| \leq \frac{1}{6n^2}$.
therefore, we need $n > 578$.

b

Prove;

The remainder of composite Simpson's rule is $E_n(f) = -\frac{(b-a)^5}{180n^4} f^{(4)}(\xi)$. $f^{(4)}(x) = e^{-x^2}(16x^4 - 48x^2 + 24)$, So $f^{(4)}(\xi) \leq 12 \Rightarrow |E_n(f)| \leq \frac{1}{15n^4}$.

Therefore, we need $n > 20$.

III

a

According to the problem statement:

$$\left\{ \begin{array}{l} \int_0^{+\infty} e^{-t}(t^2 + at + b) dx = 0, \\ \int_0^{+\infty} e^{-t}(t^2 + at + b)t dt = 0, \end{array} \right\} \Rightarrow \left\{ \begin{array}{l} 2 + a + b = 0, \\ 6 + 2a + b = 0, \end{array} \right\} \Rightarrow \left\{ \begin{array}{l} a = -4, \\ b = 2, \end{array} \right.$$

$$\text{So, } \pi_2(t) = t^2 - 4t + 2$$

b

$\pi_2(t)$ has two roots at $t = 2 - \sqrt{2}$ and $t = 2 + \sqrt{2}$.

$$\left\{ \begin{array}{l} A_1 = \int_0^{+\infty} e^{-x} \frac{x^2 - 4x + 2}{(x - 2 + \sqrt{2})(-2\sqrt{2})} dx = \frac{2 + \sqrt{2}}{4}, \\ A_2 = \int_0^{+\infty} e^{-x} \frac{x^2 - 4x + 2}{(x - 2 - \sqrt{2})(2\sqrt{2})} dx = \frac{2 - \sqrt{2}}{4}, \end{array} \right.$$

$$E_2(f) = \frac{f^{(4)}(\tau)}{4!} \int_0^{+\infty} (x^2 - 4x + 2)e^{-x} dx = \frac{f^{(4)}(\tau)}{6}$$

c

Use $I = A_1 f(t_1) + A_2 f(t_2)$, where $f(t) = \frac{1}{1+t}$ $t_1 = 2 - \sqrt{2}$ and $t_2 = 2 + \sqrt{2}$.

So, Numerically,

$$I = \frac{2 + \sqrt{2}}{4} \left(\frac{1}{3 - \sqrt{2}} \right) + \frac{2 - \sqrt{2}}{4} \left(\frac{1}{3 + \sqrt{2}} \right) \quad (13)$$

$$= \frac{4}{7} \approx 0.57142857. \quad (14)$$

$$\text{So, } E_2(f) = \frac{f^{(4)}(\tau)}{6} = \frac{4}{(1+\tau)^5} = 0.596349361 - I \approx 0.02491879.$$

IV

a

$$h_m(x_m) = 1, h_m(x_k) = 0, q_m(x_m) = 0, q_m(x_k) = 0, \quad (15)$$

$$h'_m(x_m) = 0, h'_m(x_k) = 0, q'_m(x_m) = 1, q'_m(x_k) = 0, k \neq m \quad (16)$$

$$(17)$$

$$\begin{cases} h_m(x) &= l_m^2(t)(a_m + b_mt), \\ h'_m(x) &= 2l_m(t)l'_m(t)(a_m + b_mt) + l_m^2(t)b_m, \end{cases} \Rightarrow \begin{cases} a_m + b_mx_m = 1, \\ 2l'_m(x_m)(a_m + b_mx_m) + b_m = 0, \end{cases} \Rightarrow \begin{cases} a_m = 1 + 2l'_m(x_m)x_m, \\ b_m = -2l'_m(x_m), \end{cases}$$

$$\begin{cases} q_m(x) &= l_m^2(t)(c_m + d_mt), \\ q'_m(x) &= 2l_m(t)l'_m(t)(c_m + d_mt) + l_m^2(t)d_m, \end{cases} \Rightarrow \begin{cases} c_m + d_mx_m = 0, \\ 2l'_m(x_m)(c_m + d_mx_m) + d_m = 1, \end{cases} \Rightarrow \begin{cases} c_m = -x_m, \\ d_m = 1, \end{cases}$$

b

Do Gauss intergration to find n interpolation points in the interval $[a, b]$, then do Hermite interpolation to get the interpolating polynomial $p(t) = \sum_{m=1}^n [h_m(t)f_m + q_m(t)f'_m]$. So,

$$I_n(f) = \sum_{m=1}^n [w_m f_m + \mu_m f'_m]$$

where $w_m = \int_a^b h_m(t)\rho(t) dt$ and $\mu_m = \int_a^b q_m(t)\rho(t) dt$.

c

From the discussion above, we have $\mu_m = \int_a^b q_m(t)\rho(t) dt$, if we want $\mu_m = 0$, for all $m = 1, 2, \dots, n$, we need

$$\int_a^b (-x_m + t)l_m^2(t)\rho(t) dt = 0.$$

Because $l_m(t) \in \mathbb{P}_{n-1}$, $(-x_m + t)l_m(t) \in \mathbb{P}_n$, and $(-x_m + t)l_m(t) = \prod_{k=1}^n (t - x_k)$, so when we do Gauss integration on $[a, b]$, $\prod_{k=1}^n (t - x_k)$ is orthonormal polynomial with degree n , thus $\int_a^b (-x_m + t)l_m(t)\rho(t) dt = 0$.

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Give your acknowledgements here(if any).